Learning Objectives

At the end of this lab, students will be able to:

- Understand what Data Science is.
- Understand why Python is used in Data Science.
- Install Python and Python IDLE.
- Write and execute their first Python program.
- Understand the basic structure of a Python program.
- Learn simple input and output statements.

What is Program?

A program is a sequence of instructions written in a programming language that instructs a computer to perform a specific task.

Introduction

Good Morning Students!

Welcome to the **Data Science Foundation (DSF) Laboratory**.

This is your first practical class. Don't worry if you have never written a program before. By the end of today's lab, everyone will write their first Python program successfully.

Programming is similar to learning a new language.

Just as we use English to communicate with people,

we use **Python** to communicate with computers.

What is Data Science?

Data Science is the process of collecting, analyzing, and interpreting data to make better decisions.

Real-Life Examples

- YouTube recommends videos.
- Netflix recommends movies.
- Amazon recommends products.
- Google Maps finds the fastest route.
- Swiggy predicts delivery time.

All these applications use **Data Science**.

Why Learn Python?

Python is one of the most popular programming languages because it is:

- Easy to learn
- Easy to read
- Simple syntax
- Powerful
- Free and Open Source
- Used in AI and Machine Learning
- Used in Data Analysis
- Used in Web Development
- Used in Automation

Companies Using Python

- Google
- Microsoft
- Amazon
- Netflix
- Instagram
- Facebook
- Spotify
- NASA

Applications of Python

Python is used in:

- Artificial Intelligence
- Machine Learning
- Data Science
- Web Development
- Cyber Security
- Automation
- Game Development
- Robotics

Installing Python

Download Python from

https://python.org

Install Python by checking

✔ Add Python to PATH

Then click

Next → Install

Python IDLE

IDLE means

Integrated Development and Learning Environment

It is the official editor of Python.

Two Modes of Python

1. Interactive Mode

Used for testing small commands.

Example

```
>>> 5 + 3
8
```

Advantages

- Quick testing
- Easy calculations

2. Script Mode

Used for writing complete programs.
Programs are saved with
`.py`
extension.
Example
`hello.py`

First Python Program

```
print("Hello Students")
```

Output

```
Hello Students
```

How to Run Program

1. Open IDLE
2. File
3. New File
4. Write Program
5. Save (.py)
6. Run
7. Run Module (F5)

Understanding print()

```
print("Welcome")
```

Output

```
Welcome
```

print() is used to display output on the screen.

Writing Multiple Statements

```
print("Welcome")
print("To")
print("Python")
```

Output

```
Welcome
To
Python
```

Comments in Python

Comments are ignored by Python.
They are written for humans.
Single Line Comment

```
# This is a comment
print("Hello")
```

Variables

A variable stores data.
Example

```
name = "Rahul"
age = 20
```

Rules for Variable Names

- ✔ Can contain letters
- ✔ Can contain numbers
- ✔ Can contain underscore (_)
- ✔ Cannot start with a number
- ✔ No spaces allowed

Correct

```
student_name
roll1
age
```

Incorrect

```
1name
student name
```

Input Function

Python accepts input using

```
input()
```

Example

```
name = input("Enter your name: ")
print(name)
```

Output

```
Enter your name:
Rahul
Rahul
```

Taking Two Inputs

```
name = input("Enter Name: ")
city = input("Enter City: ")

print(name)
print(city)
```

Numbers in Python

```
a = 10
b = 20
```

Addition

```
print(a+b)
```

Output

```
30
```

Common Errors Beginners Make

Forgetting Quotes

Wrong

```
print(Hello)
```

Correct

```
print("Hello")
```

Wrong Indentation

Wrong

```
print("Hello")
 print("Python")
```

Python gives

```
IndentationError
```

Missing Bracket

Wrong

```
print("Hello"
```

Python gives

```
SyntaxError
```

Hands-on Activity 1

Write a program to print

```
Welcome
To
Python Lab
```

Hands-on Activity 2

Display your information.

```
Name
Roll Number
Branch
Semester
```

Example

```
print("Name : Amit")
print("Roll : 101")
print("Branch : CSE")
print("Semester : 3")
```

Hands-on Activity 3

Take your name as input.

```
name = input("Enter Name: ")
print("Welcome", name)
```

Hands-on Activity 4

Take two numbers and print their sum.

```
a = int(input("Enter First Number: "))
b = int(input("Enter Second Number: "))

print("Sum =", a+b)
```

Lab Exercise

Program 1

Print your college name.

Program 2

Print your personal details.

Program 3

Take input from the user.

Program 4

Add two numbers.

Program 5

Multiply two numbers.

Viva Questions

1. What is Python?
2. Why is Python popular?
3. What is Data Science?
4. What is IDLE?
5. What is a variable?
6. What is the use of `print()`?
7. What is the use of `input()`?
8. Difference between Interactive Mode and Script Mode?
9. What is a comment?
10. What is the extension of a Python file?

2. What is Programming?

Ask students:

"How do you communicate with your friends?"

Students:

Using language.

Then explain:

Humans use English, Hindi, or Odia to communicate.

Computers understand only machine language (0 and 1).

Programming languages act as translators between humans and computers.

---

3. Evolution of Programming Languages

Generation	Language	Example
1GL	Machine Language	010101
2GL	Assembly Language	MOV AX, BX
3GL	High-Level Language	C, C++, Java, Python
4GL	Database Languages	SQL
5GL	AI Languages	Prolog

Explain why high-level languages are easier.

---

4. Why Python Instead of C?

This comparison helps students who have heard of C.

Feature	C	Python
Developed	1972	1991
Developer	Dennis Ritchie	Guido van Rossum
Compilation	Compiled	Interpreted
Syntax	Complex	Simple
Memory	Manual	Automatic
Lines of Code	More	Less
Beginner Friendly	Medium	Excellent
Data Science	Limited	Excellent
AI/ML	Rare	Most Popular

Example

C Program

```
#include<stdio.h>

int main()
{
printf("Hello");
return 0;
}
```

Python

```
print("Hello")
```

Students immediately understand why Python is easier.

---

5. Why Companies Prefer Python

Explain with logos (or slides).

- Google
- Netflix
- Instagram
- Spotify
- NASA
- Amazon
- Dropbox

Reason:

- Faster development
- Huge libraries
- AI support
- Data Science

---

6. Where Python is Used

Draw this on the board.

```
Python
|
├── Web Development
├── Data Science
├── AI
├── Machine Learning
├── Automation
├── Cyber Security
├── Robotics
├── IoT
├── Game Development
└── Cloud Computing
```

---

7. Python Versions

Explain:

Python 2

- Old
- No longer supported

Python 3

- Latest
- Used everywhere

Always install Python 3.

---

8. Types of IDE

Students usually know only IDLE.

Explain different IDEs.

IDE	Beginner	Professional
Python IDLE	☆☆☆☆☆	☆☆
VS Code	☆☆☆☆	☆☆☆☆☆
PyCharm	☆☆☆	☆☆☆☆☆
Spyder	☆☆☆	☆☆☆☆
Jupyter Notebook	☆☆☆☆	☆☆☆☆☆
Google Colab	☆☆☆☆☆	☆☆☆☆☆
Thonny	☆☆☆☆☆	☆☆

Explain when each is used.

---

9. IDLE vs VS Code

IDLE	VS Code
Simple	Advanced
Small	Feature-rich
Beginner	Professional
No Extensions	Thousands of Extensions
Easy Setup	Requires Configuration

Tell students:

We will start with IDLE because it helps us learn Python without distractions. Later, we'll move to VS Code for real-world development.

10. Interpreter vs Compiler

Very common viva question.

Compiler

```
Entire Program
      ↓
Machine Code
      ↓
Execution
```

Interpreter

```
Line 1 → Execute
Line 2 → Execute
Line 3 → Execute
```

Examples

Compiler

- C
- C++

Interpreter

- Python
- JavaScript

---

11. Python Features

Explain each briefly.

- Simple
- Readable
- Portable
- Open Source
- Interpreted
- Object-Oriented
- Cross Platform
- Large Community
- Huge Libraries